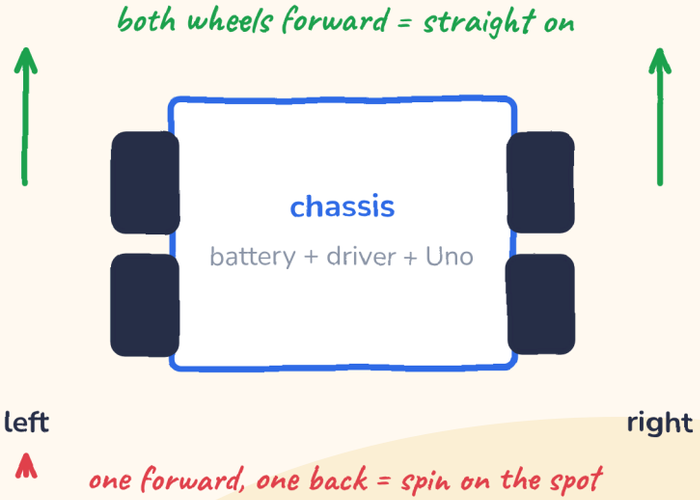
Robotics · Motors & Motion Control

Build and Wire a Robot Car

Two motors, one driver, one brain. Let's make it roll.



90 minutes | Pairs | Real kit

Did you know?

A robot with two driven wheels and one caster is called a differential drive. The same geometry steers a wheelchair, a warehouse robot and the Mars rovers' smaller cousins.

LEARN

Let's think

Why can't the Arduino just drive the motors?

The numbers that stop you

An Arduino pin can supply about **20 mA** safely, and 40 mA before it dies. A small yellow gear motor pulls **150 to 250 mA** running, and over **1 A** at the instant it starts or stalls.

That is ten to fifty times too much. The pin would not drive the motor; the motor would destroy the pin.

And there is a second problem

A pin can only be 0 V or 5 V. To run a motor backwards you must reverse the voltage across it, and a pin cannot go negative.

So we need a part that

- takes a small signal from the Arduino,
- switches a much larger current from the battery,
- and can send that current through the motor **either way round**.

That part is a **motor driver**. Inside it is an arrangement of four switches called an **H-bridge**.

LEARN

Learning aims

Today I will learn to…

1  Assemble a **2WD chassis** so it rolls straight and nothing rubs

2  Wire two motors through an **L298N driver** and a separate battery

3  Explain how an **H-bridge** reverses a motor

4  Drive **forward, back, left and right** from code, as functions

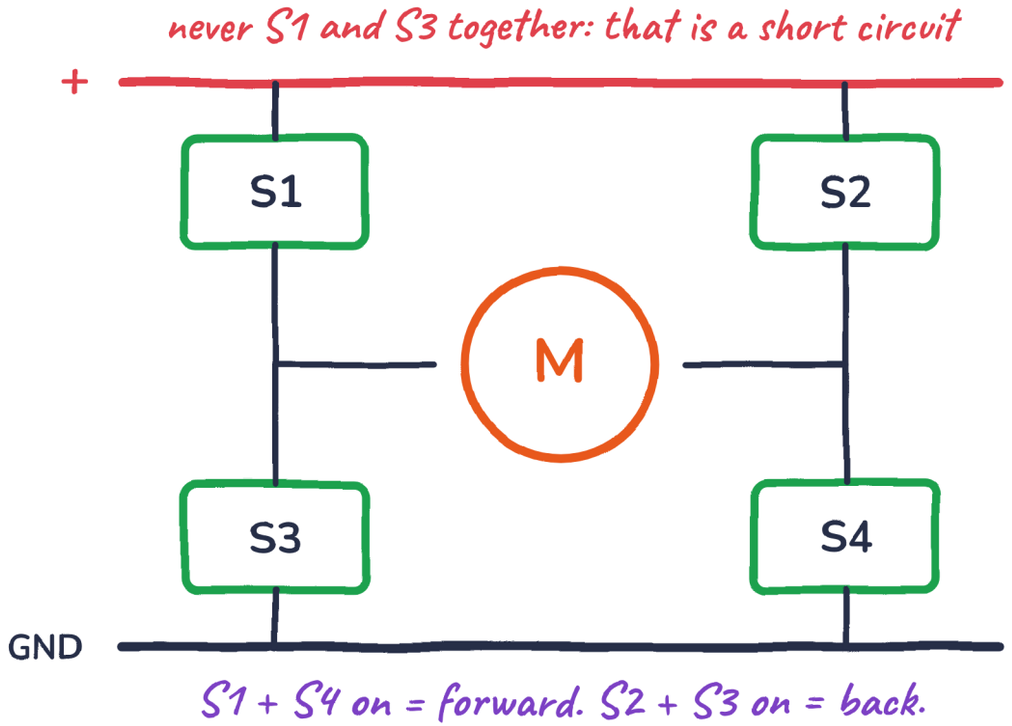
Keywords

H-bridge · differential drive · PWM · duty cycle · enable pin · stall current · common ground · caster

Objective 3 · The idea

An H-bridge is four switches

Why it is called an H



Forward

Close S1 and S4. Current enters the motor on the left and leaves on the right.

Backward

Close S2 and S3 instead. The same battery, the same motor, current the other way, so the shaft spins the other way.

Shoot-through

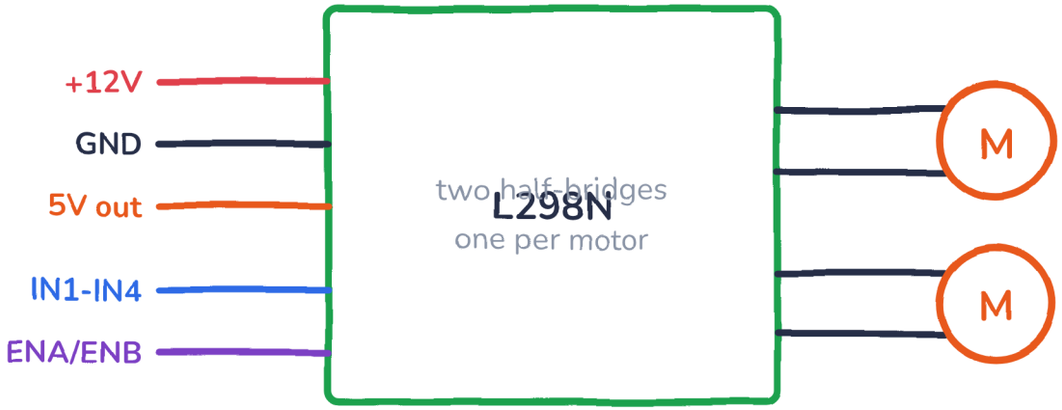
S1 and S3 together is a straight wire from + to GND. The chip gets hot and dies. This is why we let the driver do the switching, not loose wires.

LEARN

Objective 2 · The board

Reading the L298N

Two motors, one board



IN pins pick the direction. EN pins set the speed with PWM.

IN pins decide direction

IN1 HIGH and IN2 LOW drives motor A one way. Swap them and it reverses. Both the same means brake.

EN pins decide speed

ENA and ENB take PWM from pins 5, 6, 9, 10 or 11. 0 is stopped, 255 is flat out.

The 5V jumper

Leave it on and the board powers the Arduino from the battery. Take it off if you also have USB plugged in, or the two supplies fight.

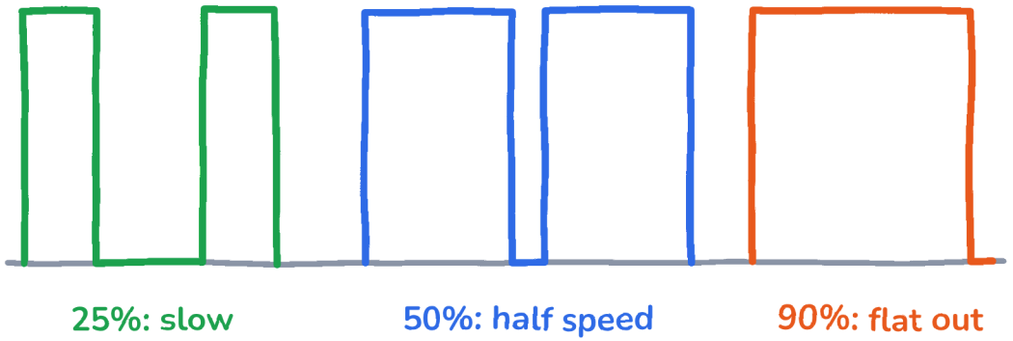
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Objective 2 · Speed

PWM: speed from a switch

Same voltage, different share of the time

The motor never gets half a volt. It gets 5 V, switched on and off fast



Why not just lower the voltage?

Because a resistor would turn the difference into heat. Switching wastes almost nothing, which matters when you are running on four AA cells.

The dead zone

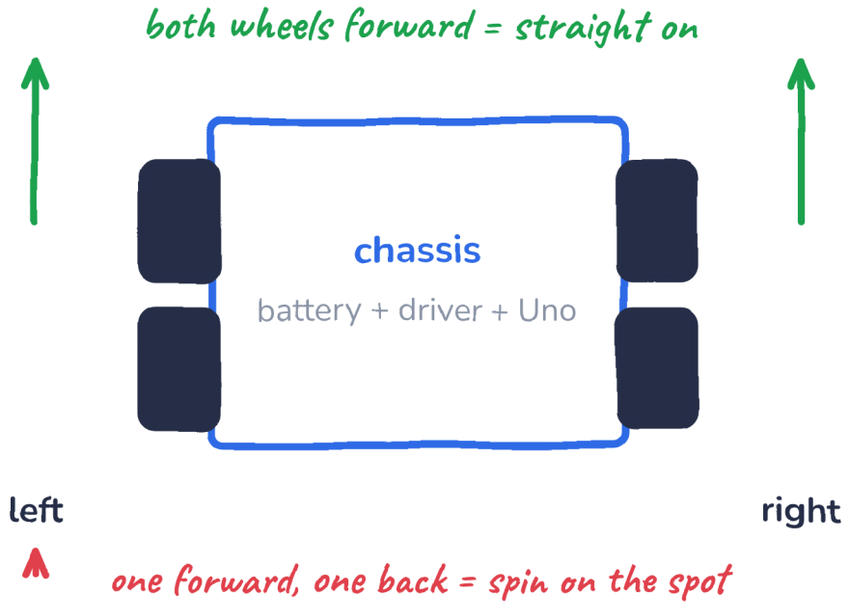
Below about **analogWrite(60)** most cheap motors only buzz. There is not enough torque to overcome their own friction. Start your speed tests at 100.

LEARN

Objective 4 · Steering

No steering wheel needed

Differential drive



Left wheel	Right wheel	The robot
forward	forward	drives straight
forward	stopped	swings a wide arc right
forward	backward	spins on the spot, right
backward	backward	reverses
fast	slow	curves gently right

In the real world

Where you have seen this

Robot vacuum

Two driven wheels, one caster, and a clever turn-on-the-spot when it meets a chair leg. Exactly this drive.

Electric wheelchair

The joystick does not steer. It sets two wheel speeds, and the chair curves because they differ.

Warehouse AGV

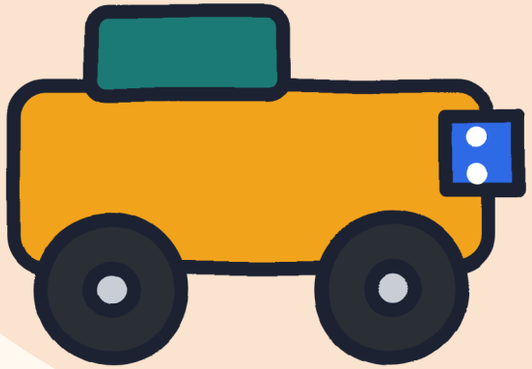
Shelf-carrying robots turn in their own footprint, because an aisle has no room for a turning circle.

Part 2 · Make it

Now let's build it

Chassis first, then wiring, then code. Test after every single step.

60 minutes | Pairs | Real kit



Before you build

What you need

Kit, per pair

- 2WD chassis kit: plate, 2 gear motors, 2 wheels, caster
- L298N motor driver
- Arduino Uno + USB cable
- 4×AA battery pack
- Male-female jumper wires, screwdriver set

Safety, before power

- Wheels **off the desk** for every first test.
- Tie long hair back. Gear motors catch it.
- Battery in last, out first, whenever you change wiring.
- If the driver smells hot, disconnect at once and tell a teacher.

MAKE

Build step 1 of 4

Assemble the chassis

Do this

1. Bolt both motors into the brackets. Do not overtighten; the plastic strips.
2. Push the wheels on until they click flat against the shaft.
3. Fit the caster at the **opposite** end to the motors.
4. Mount the battery pack over the motors, not over the caster.
5. Spin each wheel by hand. It should turn freely with no rubbing.

Why the weight goes at the back

Weight over the driven wheels gives them grip. Weight over the caster just makes the robot push a sliding lump.

Rubbing wheel?

Loosen the bracket, square the motor, retighten. A wheel that rubs will curve your robot away no matter how good your code is.

MAKE

Build step 2 of 4

Wire it up

Do this, in this order

1. Left motor to OUT1 / OUT2. Right motor to OUT3 / OUT4.
2. Battery + to **+12V**, battery – to **GND**.
3. **Arduino GND to driver GND.** Never skip this.
4. IN1→7, IN2→8, IN3→9, IN4→11.
5. ENA→5, ENB→6. Remove the little enable jumpers first.

Common ground, again

The Arduino says "5 V" compared to *its* ground. The driver measures against *its* ground. If they are not joined, the IN pins mean nothing and the robot twitches randomly.

Two supplies, one rule

Battery powers the motors. USB powers the Arduino. The grounds are shared. The 5 V rails are not.

MAKE

Build step 3 of 4

Make one wheel turn

Do this

1. Wheels off the desk.
2. Upload the sketch. The left wheel should turn forward.
3. Swap IN1 and IN2 in code. It should reverse.
4. Now repeat for the right wheel.

Wrong way round?

Do not rewrite the code. Swap the two motor wires at OUT1 and OUT2. It is the same fix, and it keeps your code honest.

```
// one motor, both directions
const int IN1 = 7, IN2 = 8, ENA = 5;

void setup() {
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENA, OUTPUT);
}

void loop() {
  digitalWrite(IN1, HIGH);     // forward
  digitalWrite(IN2, LOW);
  analogWrite(ENA, 180);       // ~70% speed
  delay(1500);

  analogWrite(ENA, 0);         // coast
  delay(500);
}
```

MAKE

Build step 4 of 4

Four functions, one robot

Do this

1. Write `forward()`, `back()`, `left()`, `right()`, `stop()`.
2. Drive a square: forward, right, four times.
3. Measure how long a 90° turn really takes. Tune it.

Why functions?

Named functions turn eight confusing digitalWrites into one word you can read out loud. Every robot you build from now on will have these five.

```
int SPD = 170;

void drive(int l, int r) {
  digitalWrite(IN1, l > 0); digitalWrite(IN2, l < 0);
  digitalWrite(IN3, r > 0); digitalWrite(IN4, r < 0);
  analogWrite(ENA, abs(l));
  analogWrite(ENB, abs(r));
}

void forward() { drive( SPD,  SPD); }
void back()    { drive(-SPD, -SPD); }
void left()    { drive(-SPD,  SPD); }
void right()   { drive( SPD, -SPD); }
void halt()    { drive(0, 0); }

void loop() {                  // a square
  for (int i = 0; i < 4; i++) {
    forward(); delay(1000);
    right();   delay(420);     // tune this
  }
  halt(); delay(3000);
}
```

Test like an engineer

Make it go straight

Symptom	Likely cause	What to try
Curves to one side	Motors are never identical	Lower the faster side: use 170 and 155, not 170 and 170
One wheel does nothing	Loose OUT screw terminal	Tug the wire gently. Retighten and retest
Both wheels twitch randomly	No common ground	Join Arduino GND to driver GND
Works on USB, dies on battery	Flat cells, or 6 V is not enough	Fresh AAs, or move to a 2S pack
Arduino resets when it starts	Motor inrush drags the rail down	Separate supplies, grounds joined

Calibration is normal

Real robots all have a trim value somewhere. Finding yours is engineering, not cheating.

LEVEL UP

Finished early?

Go further

1 

Write a **calibration routine** that drives 2 m and reports the drift

Then store the correction as a constant.

2 

Add a **soft start** so speed ramps from 0 to full over 300 ms

Less wheelspin, less voltage sag.

3 

Drive a **figure of eight** without stopping

Two different speeds, swapped halfway.

4 

Measure the **stall current** with a multimeter

Hold the wheel and watch the reading climb.

Check yourself

Can I really do it? Prove it.

1 Assemble a **2WD chassis** that rolls freely

Proof: both wheels spin by hand with no rubbing

2 Wire motors through an **L298N** with a shared ground

Proof: both motors run in both directions

3 Explain how an **H-bridge** reverses a motor

Proof: I named which two switches close for each direction

4 Drive **forward, back, left, right** as functions

Proof: my robot drove a square and came back

Rate yourself: red = not yet · amber = nearly · green = yes, and I can show someone

Show what you know

What did we learn?

Five things to keep

- Arduino pins give signals. Drivers give power.
- An H-bridge reverses a motor with four switches.
- PWM sets speed by switching fast, not by lowering volts.
- Two wheel speeds are all the steering you need.
- No shared ground, no working robot.

Exit ticket

Your robot drifts left. Name two different fixes, one mechanical and one in code.

Next class

ROB-06: servos and steppers. A DC motor spins; a servo points; a stepper counts.